A Bedside Decision Support System for Disorders of Consciousness: Multi-Scale Temporal Dynamics and Multicenter Validation

Hongjie Chen[1], Jianming Chen[1], Guanyu Xiong[2], Fang Duan[1,*], and Fang Li[3,*]

***Abstract*— Conventional bedside electroencephalography markers discard temporal sequence order, obscuring dynamic cortical reverberation critical for detecting covert consciousness. We developed an interpretable screening framework combining physically adjacent lag-one autocorrelation, normalized signal jump rate, and common-mode-decoupled residual bilateral correlation under equal-weight consensus scoring. Evaluation spanned four cohorts totaling 129 patients across recording regimes. On continuous discovery recordings from 41 patients, the primary four-feature suite attained an area under the receiver operating characteristic curve of 0.9324 under 25 cross-validation partitions, exceeding the strongest matched supervised baseline by a paired margin of 0.0360 (95 percent confidence interval 0.0090 to 0.0728). Physical sequence dependence was confirmed by 10,000 synchronous whole-channel temporal shuffles, reducing performance from 0.8911 to 0.8182 plus or minus 0.0049. On external inpatient telemetry from 15 patients at China Rehabilitation Research Center, unlabeled target standardization and direction alignment yielded an area under the curve of 0.7778, rising to 0.7963 with fixed soft truncation, compared with 0.7407 to 0.7593 for matched supervised baselines. On 120-second records from 50 patients, the jump-rate operator achieved 0.7691. Sleep-stage transitions across 23 patients provided complementary macro-scale validation over 24-hour cycles. Requiring only 2.15 milliseconds per evaluated update, this parameter-free framework provides an auditable, real-time decision support system for continuous bedside monitoring in the intensive care unit.**

***Index Terms*— Disorders of consciousness (DoC), electroencephalography (EEG), temporal dynamics, clinical decision support systems (CDSS), mesocircuit hypothesis, consensus fusion.**

I. Introduction

SEVERE acute brain injury from trauma, anoxia, or stroke leaves patients in prolonged disorders of consciousness (DoC), clinically categorized into the vegetative state/unresponsive wakefulness syndrome (VS/UWS) and the minimally conscious state (MCS) [1], [2]. Distinguishing MCS from VS/UWS informs decisions on withdrawing life-sustaining treatment, administering analgesia, and projecting rehabilitation potential. Bedside clinical evaluations carry an estimated 41% misdiagnosis rate [3]. Although standardized assessments such as the Coma Recovery Scale-Revised (CRS-R) [3] reduce diagnostic error, motor impairment and fluctuating arousal frequently confound behavioral examinations [4]. Furthermore, prospective multi-center cohorts establish that approximately 25% (and 15%–35% across multimodal cohorts) of behaviorally unresponsive patients exhibit cognitive-motor dissociation (CMD) [5], retaining volitional cortical responses without overt motor output.

Functional neuroimaging and perturbational protocols (fMRI, ^{18}F-FDG PET, TMS-EEG) [6], [7] show that preserved cortical complexity differentiates MCS from VS/UWS, but transport hazards preclude routine ICU deployment. While continuous bedside EEG provides the primary practical modality for tracking neural recovery [8]–[10], routine ICU surveillance requires sparse montages (4–8 electrodes) rather than dense research caps. Beyond static power spectra, prior investigations examined non-linear entropy [8], [11], microstates [12], sleep architecture [13], [14], long-range temporal correlations via detrended fluctuation analysis (DFA) [15], [16], source-space connectivity [17], and dynamical criticality [18]. However, clinical bedside translation remains constrained: multi-scale DFA requires stationary dense montages; source reconstruction is ill-posed on sparse arrays; entropy measures exhibit artifact sensitivity; and over-parameterized classifiers risk overfitting limited cohorts ($N < 50$). Bedside assessment has therefore relied on empirical band power, marginal dispersion ($-\mathrm{CV}_\theta$), and unadjusted bilateral correlations [10], [13], leaving non-invasive discrimination bounded to a moderate plateau (0.75–0.82 AUROC).

A. The Permutation Invariance Defect of Conventional Metrics

This diagnostic plateau stems from a shared limitation: strict permutation invariance. Conventional pipelines rely on aggregate dispersion ($-\mathrm{CV}_\theta$) and instantaneous bilateral correlation (r_{interhem}), which are invariant under temporal sequence permutation. Treating continuous EEG trajectories as unordered sets discards directional multi-second continuity and cortical reverberation.

[1]Hongjie Chen, Jianming Chen, and Fang Duan are with the College of Information Science and Engineering, Huaqiao University, Xiamen 361021, China (e-mail: 2415251001@stu.hqu.edu.cn; duan.fang@hqu.edu.cn).

[2]Guanyu Xiong is with the School of Biomedical Engineering, University of British Columbia, Vancouver, BC, Canada.

[3]Fang Li is with the Parkinson Medical Center, Beijing Rehabilitation Hospital, Capital Medical University, Beijing, China.

*Corresponding authors: Fang Duan (first; e-mail: duan.fang@hqu.edu.cn) and Fang Li (second; e-mail: fangli8608@163.com).

B. The Mesocircuit Hypothesis and Autoregressive Memory

This methodological gap diverges from established neurobiology of conscious recovery, specifically the anterior forebrain mesocircuit hypothesis [19]. Under this framework, sustained awareness depends on recurrent excitatory loops linking the central thalamus, striatum, and frontoparietal association cortices. Preserved mesocircuit connectivity manifests electrophysiologically as multi-second autoregressive continuity, quantifiable via lag-1 autocorrelation (ρ_1) across consecutive epochs. In VS/UWS, widespread deafferentation and loss of striatal inhibition disinhibit the internal globus pallidus (GPi), driving central thalamic hyperpolarization and degrading cortical activity into memoryless fluctuations. Distinguishing conscious capacity from reflexive unresponsiveness thus requires operators sensitive to physical sequence persistence.

As illustrated in Fig. 1a, individual theta trajectories in MCS (e.g., P36) exhibit sustained multi-second wave-holding, whereas VS/UWS trajectories (e.g., P25) break into decoupled transients. Static dispersion ($-\mathrm{CV}_\theta$) displays substantial overlap between cohorts ($p = 0.0264$; Fig. 1c), whereas lag-1 persistence (ρ_1) reliably separates conscious from unresponsive states ($p = 0.0227$; Fig. 1b).

While continuous telemetry enables asymptotic estimation of ρ_1, emergency bedside triage often relies on ultra-short recordings (< 5 min), where sample autocorrelation suffers elevated variance ($\mathrm{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$). For such records, we formulate the normalized jump rate (D_1) across verified adjacent epochs ($\mathbb{I}_{\text{adj}} = 1$), analytically unifying static dispersion and temporal decay via $D_1 \approx 2\pi^{-1/2}(-\mathrm{CV}_\theta)(1-\rho_1)^{1/2}$.

C. Spurious Bilateral Coupling Driven by Diffuse Slow Waves

Diffuse slow-wave volume conduction also confounds bilateral monitoring. In acute brain injury, high-amplitude delta and theta oscillations propagate broadly across the scalp via volume conduction and subcortical pacemaking. Although common average reference (CAR) subtracts linear potential offsets, rectified band-power envelopes retain substantial shared variance ($R^2 \approx 47\%$) from global pacemaking, inflating raw inter-hemispheric correlation and obscuring regional differentiation. Decoupling this common mode via ordinary least squares (OLS) orthogonal projection isolates residual bilateral resonance (r_{residual}), separating pathological co-activation in VS/UWS from preserved modular dynamics in MCS.

D. Contributions of This Work

This study introduces an interpretable bedside decision support framework with three principal contributions:

1) *Order-aware temporal operators and artifact decoupling*: We formulate lag-1 autocorrelation (ρ_1) for continuous telemetry alongside normalized adjacent jump rates (D_1) for short recordings under physical adjacency constraints ($\mathbb{I}_{\text{adj}} = 1$), analytically unified via $D_1 \approx 2\pi^{-1/2}(-\mathrm{CV}_\theta)(1-\rho_1)^{1/2}$. Synchronously shuffling 10,000 surrogate sequences degrades classification from 0.8911 AUROC to permutation baseline (0.8182 ± 0.0049, $p < 0.0001$), confirming that diagnostic separation requires chronological sequence order rather than static marginal distributions. OLS orthogonal projection decouples shared slow-wave envelope variance ($R^2 \approx 47\%$) to isolate genuine bilateral resonance (r_{residual}).

2) *Interpretable, parameter-free bedside consensus*: The parsimonious four-feature suite $\mathbf{x}_{\text{ortho}} = [c_{\text{global}}, -\mathrm{CV}_\theta, \rho_1, r_{\text{residual}}]^\top$ attains 0.9324 AUROC and 0.8998 average precision (AP) on primary discovery telemetry. Given identical inputs, equal-weight consensus scoring exceeds the top matched supervised baseline (logistic regression) by $+0.0360$ AUROC (Dirichlet paired 95% CI $[+0.0090, +0.0728]$), requiring 2.15 ms per update in cached-power replay (Supplementary Note D).

3) *Multicenter evaluation across distinct acquisition settings*: The framework is validated across 129 records spanning four independent cohorts: CRRC telemetry ($N = 15$) establishes cross-hardware generalization (0.7778 to 0.7963 AUROC); Figshare 120-second records ($N = 50$) demonstrate emergency triage utility (0.7691 AUROC); and Wielek polysomnography ($N = 23$) confirms macro-scale sleep stage transitions across circadian cycles.

II. Mathematical Formulation and Methods

Fig. 2 summarizes the processing, scoring and evaluation workflow.

A. Clinical Cohorts and Ethical Compliance

Four multicenter cohorts totaling $N = 129$ records were evaluated under institutional ethical approvals:

1) Mendeley Discovery Cohort ($N = 41$): Continuous 6-channel bedside EEG (F3, F4, C3, C4, O1, O2; 256 Hz) from 41 patients with severe brain injuries at FRCCICMR Moscow [20]: 12 MCS (8 MCS$^+$, 4 MCS$^-$; duration 6.00 ± 0.00 h) and 29 VS/UWS (5.97 ± 0.61 h), diagnosed through repeated CRS-R assessments.

2) CRRC Inpatient Cohort ($N = 15$): Prospective bedside telemetry from 15 subacute male patients (6 MCS, 9 VS/UWS; 11 ICH, 4 TBI; age 51.7 ± 12.3 years, range 28–69; duration 19.5 ± 8.0 min; Ethics No. 2025-118-01 [21]) at China Rehabilitation Research Center (CRRC) using a 32-channel NeuSen NSM2 system (1000 Hz, 4-s blocks; Supplementary Note A). Homologous 6 channels were evaluated; two patients (CRRC_008, CRRC_009) exhibited localized mechanical drift at O_2, enabling assessment on both the full cohort ($N = 15$) and the drift-free subset ($N = 13$; 6 MCS including 1 EMCS, 7 VS/UWS).

3) Figshare Short-Record Analysis ($N = 50$): High-density resting-state EEG (250 Hz, 120-s records) from 50 patients with prolonged DoC (19 MCS, 31 VS/UWS) at Hangzhou Mingzhou Brain Rehabilitation Hospital [12], [22], evaluating rapid bedside triage using homologous channels under fixed short-duration recordings.

4) Wielek PSG Cohort ($N = 23$): 24-hour polysomnography from 23 post-comatose patients (11 MCS, 12 VS/UWS) [13], with 30-second hypnograms (Wake, N1, N2, N3, REM) tracking macro-scale transition continuity across circadian sleep cycles.

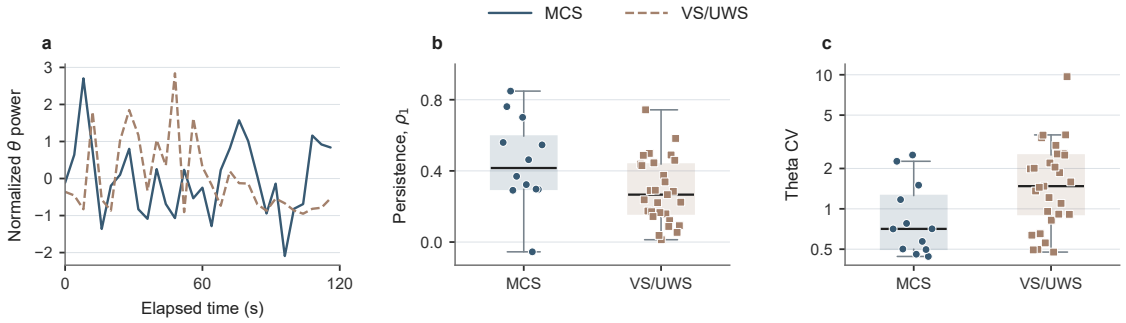


Fig. 1. Bedside EEG dynamics and patient distributions. (a) Standardized theta power across 30 consecutive 4-s epochs in P36 (MCS) and P25 (VS/UWS). (b,c) Lag-1 persistence $\boldsymbol{\rho_1}$ ($\boldsymbol{p = 0.0227}$) and positive theta CV ($\boldsymbol{p = 0.0264}$) across 12 MCS and 29 VS/UWS patients (two-sided Mann–Whitney tests). Points show individual patients; boxes show medians and IQRs; whiskers span 1.5 IQR. Panel c uses a logarithmic scale.

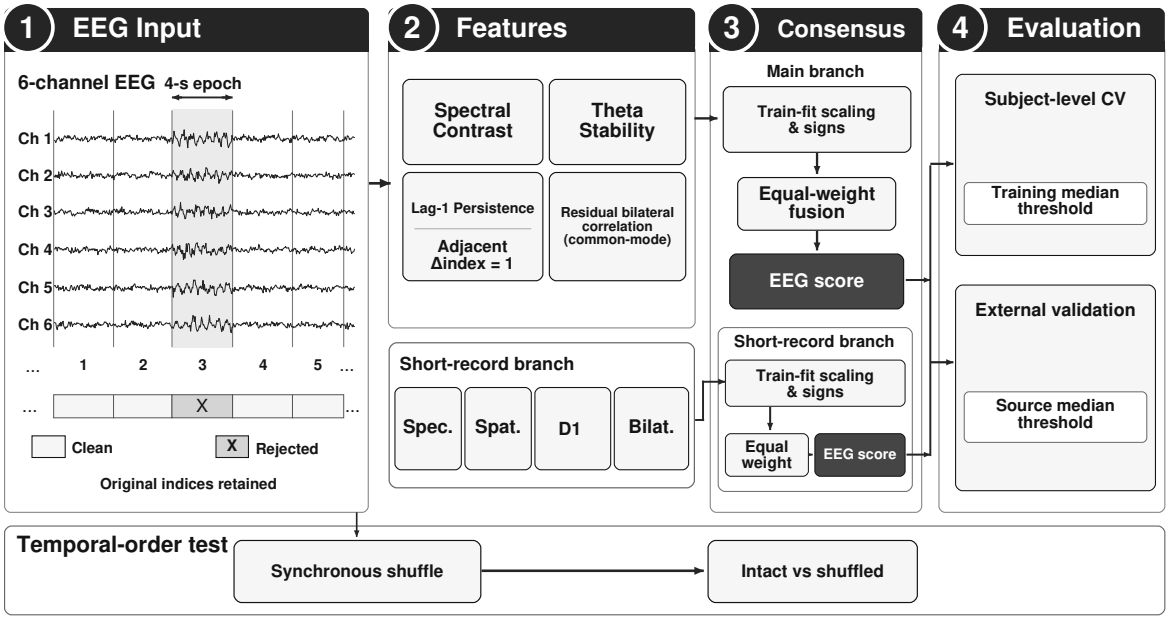


Fig. 2. System overview. Schematic EEG traces illustrate processing. Original block indices are retained, and lag-1 persistence and D1 use only physically adjacent clean blocks ($\boldsymbol{\Delta \mathrm{index} = 1}$). Main and short-record branches use distinct four-feature suites and independently fitted equal-weight consensus scores. In the short-record branch, Spec., Spat., and Bilat. denote spectral contrast, spatial contrast, and bilateral correlation. Discovery uses 4-s epochs; Figshare triage uses 2-s segments. Source training data determine the operating median. External cohorts include target-batch standardization and direction alignment where indicated. The temporal-order test compares intact data with 10,000 synchronized multichannel permutations using the frozen intact scale.

B. EEG Acquisition, Preprocessing, and Physical Adjacency Constraint

Continuous 6-channel EEG (256 Hz) was re-referenced to Common Average Reference (CAR) and segmented into non-overlapping 4-second epochs (1,024 samples). Spectral densities were computed across each epoch via Hann-tapered periodograms ($\Delta f = 0.25$ Hz), extracting theta (4–8 Hz) and alpha (8–13 Hz) band power.

Artifact screening at $500\,\mu$V retained $5{,}185.5 \pm 595.6$ blocks per patient ($96.98\% \pm 3.2\%$, 5.76 ± 0.66 h; rejection $2.84\% \pm 2.1\%$ in MCS vs. $3.09\% \pm 3.6\%$ in VS/UWS, $p = 0.82$, Mann–Whitney U test). A conservative $150\,\mu$V cutoff retained $90.14\% \pm 17.87\%$ blocks ($4{,}672.9 \pm 1{,}024.7$ pairs; minimum 668), confirming balanced retention.

Let $\mathcal{T} = \{t_1, t_2, \ldots, t_N\}$ denote clean epoch indices in a continuous session, and let G_{t_i} denote spatial mean theta power across all 6 channels at epoch t_i. To prevent spurious concatenation across artifact gaps, the physical adjacency indicator is defined as:

$$\mathbb{I}_{\text{adj}}(t_i, t_{i+1}) = \begin{cases} 1, & \text{if } t_{i+1} - t_i = 1, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

where 0 denotes an intervening artifact gap, and $\mathbb{I}_{\text{adj}} = 1$ denotes consecutive epochs. Sequential metrics operate strictly on pairs with $\mathbb{I}_{\text{adj}} = 1$, retaining $M = 5{,}133.3 \pm 687.7$ adjacent pairs per patient (minimum 1,678, exceeding the validity threshold $M \geq 10$).

C. Canonical Static Features and Order-Aware Dynamics

Four baseline features define $\mathbf{x}_{\text{canonical}} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, r_{\text{interhem}}]^\top \in \mathbb{R}^4$. We evaluate three expanded feature spaces: the 5-feature temporal extension $\mathbf{x}_{+\rho_1} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, r_{\text{interhem}}, \rho_1]^\top \in \mathbb{R}^5$; the 5-feature combined suite $\mathbf{x}_{\text{combined}} = [c_{\text{global}}, c_{\text{spatial}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]^\top \in \mathbb{R}^5$; and the primary parsimonious 4-feature deployment suite $\mathbf{x}_{\text{ortho}} = [c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]^\top \in \mathbb{R}^4$. Theta power (4–

TABLE I

PERFORMANCE OF ORDER-AWARE TEMPORAL DYNAMICS, CONSENSUS FUSION, AND MACHINE LEARNING BASELINES ON MENDELEY COHORT (25 OUTER PARTITIONS; 95% CIS PAIRED VS. CANONICAL LINEAR BASELINE VIA 10,000 DIRICHLET RESAMPLES; PR-AUC REPORTED AS AVERAGE PRECISION [AP]; MATCHED SUPERVISED ROWS USE THE SAME FOUR FEATURES AS THE DEPLOYMENT SUITE; 5-FEAT COMBINED MODELS SUBSTITUTE r_{INTERHEM} WITH r_{RES}).

Model / Feature Configuration	AUROC	PR-AUC	Balanced Acc.	Sensitivity	Specificity	95% CI vs. Baseline
Consensus Aggregation Frameworks						
Canonical Static (4 Feats, Linear)	0.8200	0.7823	0.7201	0.8333	0.6069	Baseline
Primary Orthogonal (4 Feats, Linear)	**0.9324**	**0.8998**	**0.8101**	0.9167	0.7034	$[+0.0174, +0.2186]$
Primary Orthogonal (4 Feats, Rank Consensus)	**0.9076**	**0.8548**	**0.8448**	1.0000	0.6897	$[+0.0137, +0.1725]$
Plus Lag-1 ρ_1 (5 Feats, Linear)	0.8911	0.8384	0.7997	0.9167	0.6828	$[+0.0011, +0.1534]$
Plus $\rho_1 + r_{\text{res}}$ (5 Feats, Linear, $c = 1.5\sigma$)	**0.9149**	**0.8782**	**0.8399**	0.9833	0.6966	$[+0.0064, +0.1931]$
Plus $\rho_1 + r_{\text{res}}$ (5 Feats, Rank Consensus)	0.8949	0.8353	0.7741	0.9000	0.6483	$[+0.0049, +0.1530]$
T-GECF Adaptive Gated Hybrid (5 Feats)	**0.9116**	**0.8669**	**0.8399**	0.9833	0.6966	$[+0.0013, +0.1927]$
Supervised & Domain Baseline Benchmarks						
Lempel-Ziv Complexity (LZC)	0.5136	0.5061	0.5267	0.5500	0.5034	$[-0.4233, -0.1862]$
Shannon Spectral Entropy (SE)	0.6953	0.6182	0.6721	0.7167	0.6276	$[-0.2540, +0.0085]$
Canonical Static (Tuned Ridge)	0.7653	0.7193	0.6661	0.7667	0.5655	$[-0.0998, -0.0140]$
Orthogonal 4-Feat (Tuned Ridge)	0.8813	0.8393	0.7825	0.9167	0.6483	$[-0.0446, +0.1744]$
Canonical Static (Tuned Logistic)	0.7722	0.7236	0.6917	0.7833	0.6000	$[-0.0959, -0.0064]$
Orthogonal 4-Feat (Tuned Logistic)	0.8964	0.8524	0.8080	0.9333	0.6828	$[-0.0225, +0.1826]$
Canonical Static (Shrinkage-LDA)	0.7709	0.7252	0.6626	0.7667	0.5586	$[-0.0884, -0.0162]$
Orthogonal 4-Feat (Shrinkage-LDA)	0.8769	0.8424	0.7859	0.9167	0.6552	$[-0.0496, +0.1699]$
Representation Ablations & Operator Variants						
Canonical Static (4 Feats, Rank Consensus)	0.7893	0.7299	0.7270	0.8333	0.6207	$[-0.0619, -0.0076]$
Plus Lag-1 ρ_1 (5 Feats, Rank Consensus)	0.8591	0.8068	0.7187	0.8167	0.6207	$[-0.0188, +0.1053]$
Plus Jump D_1 (5 Feats, Linear)	0.8611	0.8012	0.7491	0.8500	0.6483	$[-0.0018, +0.0911]$
Plus Jump D_1 (5 Feats, Rank Consensus)	0.8424	0.7787	0.7132	0.8333	0.5931	$[-0.0208, +0.0694]$
Replace $-\text{CV}_\theta$ with ρ_1 (Linear)	0.8851	0.8495	0.7678	0.8667	0.6690	$[-0.0223, +0.1613]$
Replace $-\text{CV}_\theta$ with D_1 (Linear)	0.8720	0.8097	0.7776	0.9000	0.6552	$[+0.0050, +0.1094]$
Residual Correlation r_{res} (Linear)	0.8433	0.7671	0.7575	0.8667	0.6483	$[-0.0561, +0.1070]$

8 Hz) is the primary carrier because thalamocortical mesocircuit projections drive frontoparietal dynamics within this band [19], whereas delta reflects non-specific slowing and alpha is largely abolished in VS/UWS. Let $P_{\theta,t}^{(k)}$ and $P_{\alpha,t}^{(k)}$ denote theta and alpha powers of channel $k \in \{1, \ldots, 6\}$ at clean epoch $t \in \{1, \ldots, N\}$.

1) Global Spectral Contrast (c_{global}): Quantifies the ratio of slow to fast power across all channels: $c_{\text{global}} = (\bar{P}_\theta - \bar{P}_\alpha)/(\bar{P}_\theta + \bar{P}_\alpha + \epsilon)$, where $\bar{P}_\theta = \frac{1}{6N}\sum_{k,t} P_{\theta,t}^{(k)}$, $\bar{P}_\alpha = \frac{1}{6N}\sum_{k,t} P_{\alpha,t}^{(k)}$, and $\epsilon = 10^{-12}$.

2) Spatial Contrast (c_{spatial}): Computes mean slow-to-fast ratio: $c_{\text{spatial}} = \frac{1}{6}\sum_{k=1}^{6}(\bar{P}_\theta^{(k)} - \bar{P}_\alpha^{(k)})/(\bar{P}_\theta^{(k)} + \bar{P}_\alpha^{(k)} + \epsilon)$. Fixed feature configurations were compared to assess spectral redundancy; the four-feature suite omitted spatial contrast (Section III-C).

3) Negative Theta Dispersion ($-\text{CV}_\theta$): Measures temporal variance of mean theta $G_t = \frac{1}{6}\sum_{k=1}^{6} P_{\theta,t}^{(k)}$: $-\text{CV}_\theta = -\sqrt{\frac{1}{N}\sum_{t=1}^{N}(G_t - \bar{G})^2}/(\bar{G} + \epsilon)$, with temporal mean $\bar{G} = \frac{1}{N}\sum_{t=1}^{N} G_t$.

4) Inter-Hemispheric Bilateral Correlation (r_{interhem}): Averages raw pairwise Pearson correlations across symmetric bilateral pairs: $r_{\text{interhem}} = \frac{1}{3}\sum_{(L,R)} \text{Corr}(\mathbf{y}_L, \mathbf{y}_R)$, where $\mathbf{y}_L, \mathbf{y}_R \in \mathbb{R}^N$ denote the 4-second block theta-power vectors across symmetric pairs $(L, R) \in \{(\text{F3}, \text{F4}), (\text{C3}, \text{C4}), (\text{O1}, \text{O2})\}$.

To capture multi-second sequential continuity, two order-aware temporal representations are formulated on physically adjacent blocks:

Lag-1 Memory Persistence (ρ_1): Quantifies autoregressive temporal persistence across consecutive 4-second blocks:

$$\rho_1 = \text{Corr}(\{G_{t_i} \mid \mathbb{I}_{\text{adj}} = 1\}, \{G_{t_{i+1}} \mid \mathbb{I}_{\text{adj}} = 1\}). \quad (2)$$

High ρ_1 values reflect smooth baseline preservation, whereas low values signify rapid temporal decorrelation.

Normalized Adjacent Jump Rate (D_1): Measures resistance to high-frequency baseline discontinuities:

$$D_1 = -\frac{1}{M}\sum_{i:\mathbb{I}_{\text{adj}}=1} |G_{t_{i+1}} - G_{t_i}|/(\bar{G} + \epsilon), \quad (3)$$

where $\bar{G} = \frac{1}{N}\sum_{t=1}^{N} G_t$, $M = \sum \mathbb{I}_{\text{adj}}$, and $\epsilon = 10^{-12}$. Under stationary Gaussian increments $\Delta G_t \sim \mathcal{N}(0, 2\sigma^2(1-\rho_1))$, expected jump magnitude satisfies $\mathbb{E}[|\Delta G_t|] = 2\pi^{-1/2}\sigma(1-\rho_1)^{1/2}$, yielding $D_1 \approx 2\pi^{-1/2}(-\text{CV}_\theta)(1-\rho_1)^{1/2}$. Evaluating D_1 on log-power increments accounts for 93.91% of empirical variance under this identity (Section III-C). Given estimator variance $\text{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$, D_1 provides a low-variance proxy for brief records ($M < M^* \approx 75$ epochs), whereas ρ_1 achieves asymptotic efficiency.

To test macro-scale persistence on the Wielek PSG cohort ($N = 23$), stage stay persistence on 30-s hypnograms $\{s_t\}_{t=1}^{L} \in \{\text{Wake, N1, N2, N3, REM}\}$ is $P_{\text{stay}} = \frac{1}{L-1}\sum_{t=1}^{L-1}\mathbb{I}(s_{t+1} = s_t)$, with transition flexibility $T_{\text{trans}} = 1 - P_{\text{stay}}$.

Together, these features form a hierarchical evaluation system. The four-dimensional feature space is spanned by four complementary physical domains: spectral slowing contrast, marginal envelope dispersion, sequential autoregressive continuity, and residual bilateral differentiation. The parsimonious 4-feature suite $[c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]$ is the primary clinical deployment system, while orthogonal factorials and channel ablations systematically verify the independent contribution of each physical operator.

D. Residual Bilateral Resonance (r_{residual})

For each symmetric bilateral pair $(L, R) \in \{(\text{F3}, \text{F4}), (\text{C3}, \text{C4}), (\text{O1}, \text{O2})\}$ with block theta-power vectors $\mathbf{y}_L, \mathbf{y}_R \in \mathbb{R}^N$, the nuisance signal across the remaining 4 channels is $z_t^{(L,R)} = \frac{1}{4}\sum_{k \notin \{L,R\}} P_{\theta,t}^{(k)}$. With standardized $\tilde{\mathbf{z}}^{(L,R)} \in \mathbb{R}^N$ and design matrix $\mathbf{X}_c^{(L,R)} = [\mathbf{1}, \tilde{\mathbf{z}}^{(L,R)}] \in \mathbb{R}^{N \times 2}$,

TABLE II
MULTISCALE SENSITIVITY AND EXTERNAL COHORT BENCHMARKS

(A) Multiscale Window & Cohort Transfer (AUROC)		
Cohort / Duration	**Linear**	**Rank Consensus**
Mendeley 2-second	0.8916	0.8529
Mendeley 4-second (Default)	**0.8911**	**0.8591**
Mendeley 8-second	0.8711	0.8453
Mendeley 30-min Trunc.	0.8867	0.8480
Figshare Static Baseline	0.7199	0.6812
Figshare 120-s ($N = 50$, Jump D_1)	**0.7691**	0.6927
Figshare 120-s ($N = 50$, Lag-1 ρ_1)	0.7012	0.6845
Figshare 25-Fold CV (D_1, $N = 50$)	**0.7771**	0.7505
(B) Longitudinal & Multi-Tier Benchmarks		
Validation Metric		**Value**
Mendeley 5-h ICC (ρ_1 / D_1 / r_{residual})		0.4857 / 0.6433 / 0.4511
Wielek Transition AUC (directional $p = 0.030$)		**0.7348**
Wielek Mean State Stay (MCS vs. VS)		0.7393 vs. 0.8361
Figshare Zero-Shot BA (Train Thresh.)		0.5645 (Sens 1.000)
62-Channel HD Mode-1 R^2		46.52%

the exact unregularized OLS projection operator $\mathbf{P}^{\perp}_{(L,R)} = \mathbf{I}_N - \mathbf{X}_c^{(L,R)}(\mathbf{X}_c^{(L,R)})^{+} \in \mathbb{R}^{N\times N}$ (with $\operatorname{rank}(\mathbf{P}^{\perp}) = N - \operatorname{rank}(\mathbf{X}_c)$, where $\mathbf{X}_c^{+}$ denotes the Moore–Penrose pseudoinverse) computes orthogonalized residuals $\boldsymbol{\epsilon}_L^{(L,R)} = \mathbf{P}^{\perp}_{(L,R)}\mathbf{y}_L$ and $\boldsymbol{\epsilon}_R^{(L,R)} = \mathbf{P}^{\perp}_{(L,R)}\mathbf{y}_R$. The residual bilateral resonance averages pairwise correlations:

$$r_{\text{residual}} = \frac{1}{3}\sum_{(L,R)} \operatorname{Corr}(\boldsymbol{\epsilon}_L^{(L,R)}, \boldsymbol{\epsilon}_R^{(L,R)}). \tag{4}$$

Although CAR eliminates instantaneous potential offsets, rectified theta power envelopes retain shared slow-wave variance ($\approx 47\%$). Decoupling this common mode via OLS projection isolates intrinsic residual bilateral coupling across homologous pairs.

E. Consensus Operators and Machine Learning Baselines

Linear Standardized Aggregation (Linear): Equal-weight consensus follows the fixed-coefficient formulation of Dawes improper linear models [23]. The score is $S_{\text{linear}}(\mathbf{x}) = K^{-1}\sum_{k=1}^{K} s_k z_k$, where $z_k = (x_k - \mu_k^{\text{tr}})/\sigma_k^{\text{tr}}$ and each sign s_k is fitted from the feature–label correlation in the source training partition. Standardization and signs are re-estimated inside every inner and outer training fold. For the full-source four-feature fit, the signs are $[+1, +1, +1, -1]^{\top}$. CRRC target alignment uses the specified vector $[-1, +1, +1, +1]^{\top}$ for all models in common oriented coordinates (Supplementary Note C). A separate soft-truncated score is $K^{-1}\sum_k \operatorname{clip}(s_k z_k, -c, c)$ with fixed $c = 1.5$. The primary four-feature model uses $c = \infty$; the five-feature truncation grid and four-feature truncated variant are reported separately.

Non-Parametric Rank Consensus (S_{rank}): Adapted from multi-criteria decision frameworks [24], rank consensus maps signed features to empirical cumulative distribution function (CDF) midranks:

$$S_{\text{rank}}(\mathbf{x}) = \frac{1}{K}\sum_{k=1}^{K} \tilde{F}_k(s_k x_k), \tag{5}$$

where $\tilde{F}_k(u) = \frac{1}{N_{\text{tr}}}\sum_{i=1}^{N_{\text{tr}}}[\mathbb{I}(s_k x_{i,k}^{\text{tr}} < u) + \frac{1}{2}\mathbb{I}(s_k x_{i,k}^{\text{tr}} = u)]$. Curvature is fixed to $p = 1.0$ (mean on ranks), ensuring constant marginal substitution without uncalibrated parameters on small cohorts.

Adaptive Fluctuation-Gated Hybrid (T-GECF): To combine linear sensitivity with rank stability under sensor disturbances, T-GECF couples the two estimators via sample-level volatility gating. For signed standardized features $\tilde{x}_k = s_k(x_k - \mu_k^{\text{tr}})/(\sigma_k^{\text{tr}} + \epsilon)$, sample volatility is $v(\mathbf{x}) = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(\tilde{x}_k - \bar{\tilde{x}})^2}$, standardized to $z_v(\mathbf{x}) = (v(\mathbf{x}) - \mu_v^{\text{tr}})/(\sigma_v^{\text{tr}} + \epsilon)$ against training folds. A continuous Sigmoid gate $\beta(\mathbf{x}) = 1/(1 + \exp(-\kappa(z_v(\mathbf{x}) - \theta_0)))$ ($\theta_0 = 1.645, \kappa = 2.0$) mixes standardized rank and linear scores:

$$S_{\text{T-GECF}}(\mathbf{x}) = \beta(\mathbf{x})\tilde{S}_{\text{rank}}(\mathbf{x}) + (1 - \beta(\mathbf{x}))\tilde{S}_{\text{linear}}(\mathbf{x}), \tag{6}$$

where $\tilde{S}_{\text{rank}}$ and $\tilde{S}_{\text{linear}}$ are standardized to zero mean and unit variance using outer fold statistics prior to mixing. Elevated volatility ($z_v(\mathbf{x}) > 1.645$) dynamically shunts decision weight toward bounded rank consensus $\tilde{S}_{\text{rank}}(\mathbf{x})$, mitigating the influence of isolated channel artifacts or amplitude spikes.

Supervised Machine Learning and Domain Baselines: Ridge, logistic regression, and shrinkage LDA receive exactly the same feature columns, standardized coordinates, direction coding, and subject partitions as equal-weight scoring. Ridge α and logistic C are selected from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ using source-only inner three-fold stratified cross-validation; transforms are fitted on each inner training fold. LDA uses analytical shrinkage (`auto`) [25]. The same source-selected parameters are retained across external protocols. The primary matched comparison uses $\mathbf{x}_{\text{ortho}}$; the short-record branch uses $[c_{\text{global}}, c_{\text{spatial}}, D_1, r_{\text{interhem}}]^{\top}$ for every model. Lempel-Ziv complexity, spectral entropy, and an EEGNet-style CNN [26] provide additional domain baselines under the recorded protocols.

F. Subject-Level Validation Protocol and Decision Thresholds

Discovery evaluation uses five random seeds and five stratified subject-level folds per seed, giving 25 frozen outer partitions. Source means, scales, signs, empirical ranks, and hyperparameters are fitted strictly within training partitions. AUROC and average precision (AP) are averaged across outer folds; sensitivity, specificity, balanced accuracy, and F1 use pooled outer-test decisions, with each subject contributing once per seed. Each model's operating threshold is its source-training score median. Paired 95% intervals use 10,000 subject-level Dirichlet Bayesian bootstrap draws [27], sharing each subject's weight across folds and methods conditional on the fitted predictions. External validation separates frozen-source transfer, unlabeled target-batch standardization, and CRRC target direction alignment. Models share the same target transform within each protocol, remaining blinded to external labels until final metric evaluation. All reported classification operating points use source-training medians. Full parameters, transforms, thresholds, and predictions are archived with the evaluation scripts. A multivariable logistic regression on standardized $[c_{\text{global}}, R^2, \rho_1]$ provides the conditional association analysis.

III. EXPERIMENTAL RESULTS

A. The 2×2 Orthogonal Factorial Decomposition

Table I reports classification metrics across model and feature configurations. The 2×2 factorial decomposition

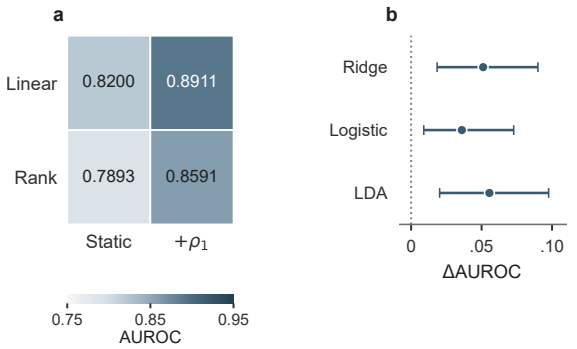


Fig. 3. Representation and scoring-rule comparisons in the 41-patient discovery cohort. (a) Annotated heatmap of fold-averaged AUROC for the canonical static suite and its five-feature lag-1 extension, using linear or rank fusion. Colors follow the displayed AUROC scale. (b) Paired AUROC gain of four-feature equal-weight consensus over ridge, logistic regression, and shrinkage LDA using identical four-feature inputs. Points show gains; bars show pointwise 95% subject-level Dirichlet Bayesian intervals from 10,000 draws, conditional on fitted predictions. The dotted line denotes zero gain.

(Fig. 3a) yields four anchor quadrants: Cell A (Static Baseline, Linear) achieves AUROC = 0.8200; Cell B (Static Baseline, Rank) yields 0.7893; Cell C (Plus Lag-1 ρ_1, Linear) attains 0.8911, the highest discrimination among the four factorial configurations, whereas Cell D (Plus Lag-1 ρ_1, Rank) achieves 0.8591. The temporal representation net gain is $\Delta_{\text{rep}} = \text{AUC(Cell C)} - \text{AUC(Cell A)} = +0.0711$ (95% CI $[+0.0011, +0.1534]$, probability of positive gain 97.74%). Linear and rank consensus address complementary objectives ($\Delta_{\text{fuse}} = \text{AUC(Cell D)} - \text{AUC(Cell C)} = -0.0320$): linear aggregation preserves continuous numerical spread, whereas rank consensus provides bounded stability against outliers.

B. Diagnostic Operating Characteristics and Imbalanced Precision-Recall

At the pre-specified training median threshold, incorporating ρ_1 elevated Balanced Accuracy from 0.7201 to 0.7997 and Sensitivity from 0.8333 to 0.9167 (Table I). Combining lag-1 persistence with residual bilateral resonance ($\mathbf{x}_{\text{combined}}$) improved the five-feature extension: under soft truncation ($c = 1.5\sigma$), linear aggregation achieved **0.9149** AUROC (PR-AUC = **0.8782**, Balanced Accuracy = **0.8399**, Sensitivity = 0.9833). In rank consensus, this suite elevated AUROC to **0.8949**, matching unconstrained linear aggregation within 0.0047 (0.8996).

Paired Bayesian bootstrap intervals for rank consensus, soft truncation, and T-GECF remain strictly positive ($[+0.0049, +0.1530]$, $[+0.0064, +0.1931]$, and $[+0.0013, +0.1927]$ vs. Canonical Static Baseline 0.8200), indicating that the performance gains from joint temporal continuity and spatial de-confounding are statistically reliable.

Stratifying the 12 MCS patients by behavioral subdiagnosis revealed consistent temporal persistence across subtypes: MCS$^+$ patients ($N = 8$, characterized by command following or intelligible verbalization) showed $\rho_1 = 0.4563 \pm 0.2687$ and were correctly identified in 8 of 8 cases (100% sensitivity), while MCS$^-$ patients ($N = 4$, demonstrating visual pursuit or localization only) exhibited $\rho_1 = 0.4377 \pm 0.1670$ with 3 of 4 cases correctly classified (75% sensitivity; overall MCS sensitivity 11/12 = 91.67%). Under class imbalance (12 MCS vs. 29 VS/UWS, prevalence baseline PR-AUC $= 0.2927$), incorporating ρ_1 advanced PR-AUC from 0.7823 (Cell A) to **0.8384** (Cell C), with direct substitution (Replace CV by ρ_1) reaching **0.8495** and the combined model reaching **0.8782**, indicating diagnostic stability under skewed class priors.

C. Feature Ablation and Collinearity Redundancy Analysis

Directly replacing $-\text{CV}_\theta$ with order-aware ρ_1 within the 4-feature suite yielded 0.8851 AUROC, showing that sequence memory supersedes marginal variance. Across 41 patients, MCS exhibited higher lag-1 persistence (0.450 ± 0.250 vs. 0.287 ± 0.181, $p = 0.0227$, AUROC = 0.7299; Fig. 1b,c) and attenuated normalized adjacent jump rates (D_1: -0.420 ± 0.089 vs. -0.558 ± 0.168, $p = 0.0112$, AUROC = 0.7557). Evaluating jump rate D_1 on log-transformed power increments empirically verifies the analytical identity across subjects ($r = 0.9691, p = 2.59 \times 10^{-25}, R^2 = 93.91\%$). Because estimator variance scales as $\text{Var}(\hat{\rho}_1) \sim \mathcal{O}(1/M)$, D_1 provides a stable proxy below the crossover horizon $M^* \approx 75$ epochs (5 min) before asymptotic autoregressive efficiency dominates. Central sensorimotor channels (C3–C4) overlying thalamocortical pathways exhibited highest univariate discrimination (ρ_1: 0.3996 ± 0.1863 in MCS vs. 0.2611 ± 0.1631 in VS/UWS, $p = 0.0132$, AUROC = 0.7500), outperforming frontal (0.6264) and occipital (0.6810) pairs.

Regressing out the global common mode (r_{residual}) improved AUROC from 0.8200 to 0.8433, accounting for $R^2 = 47.27\% \pm 18.43\%$ of raw bilateral variance uniformly across groups (46.45% in MCS vs. 47.61% in VS/UWS; $p = 0.7418$). Whereas raw bilateral correlation was non-significant (0.563 ± 0.168 in MCS vs. 0.661 ± 0.146 in VS/UWS, $p = 0.0605$, AUROC = 0.6897), residualization unmasked residual bilateral differentiation beyond the shared background power envelope (0.233 ± 0.080 vs. 0.376 ± 0.193, $p = 0.0211$, AUROC = 0.7328), reflecting modularity in MCS versus hyper-synchrony in VS/UWS. Multivariable logistic regression controlling for c_{global} ($z = -0.812, p = 0.417$) and common-mode R^2 ($z = 0.329, p = 0.742$) confirmed that ρ_1 provides significant independent diagnostic value ($z = 2.460, p = 0.0139$, adjusted OR = 4.260, 95% CI $[1.343, 13.512]$), confirming conditional independence.

Because c_{global} and c_{spatial} display high empirical collinearity ($r = 0.9813, p = 1.58 \times 10^{-29}$), we evaluated its omission as a fixed representation ablation. In the redundancy ablation (Table I), removing c_{spatial} to form the 4-feature deployment suite $[c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]$ elevated linear AUROC to **0.9324** (PR-AUC = **0.8998**; Balanced Accuracy = 0.8101, 95% CI $[+0.0174, +0.2186]$ vs. baseline, positive gain probability 99.16%) and raised rank consensus to **0.9076** (95% CI $[+0.0137, +0.1725]$, sensitivity 1.0000). Similarly, dropping c_{global} reached 0.9289 AUROC in linear aggregation and 0.9082 in rank consensus. The four-feature configuration is the primary deployment suite. The five-feature truncated suite (0.9149) provides a complementary ablation for assessing redundancy and bounded-influence scoring.

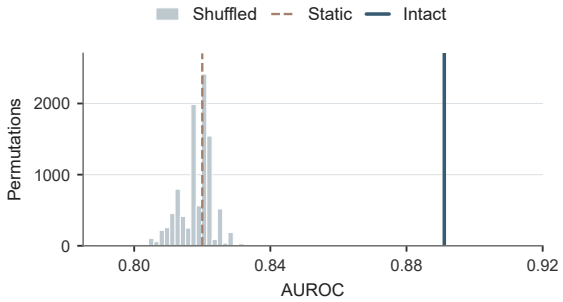


Fig. 4. Frequency distribution of AUROC under 10,000 synchronous time shuffles for the 5-feature lag-1 extension with intact scales (30 equal-width bins). Shuffled AUROC is $\mathbf{0.8182 \pm 0.0049}$ (mean $\pm$ SD); none reached intact AUROC 0.8911 ($\boldsymbol{p < 0.0001}$). Vertical lines mark static baseline (0.8200, dashed) and intact score (solid).

D. Temporal-Order Dependency via 10,000 Synchronous Surrogates

To verify that performance gains originate strictly from temporal sequence order rather than incidental distributional properties, 10,000 synchronous whole-montage permutation surrogates were evaluated across all 41 patients [28]. Freezing intact training scales ($\sigma_k^{\text{tr, intact}}$) prevents inverse noise amplification on attenuated shuffled variances. Disrupting chronological sequence alignment degraded macro AUROC across all temporal configurations (Fig. 4): Plus Lag-1 (Linear) dropped from 0.8911 to 0.8182 ± 0.0049 (range: $[0.7902, 0.8369]$, empirical $p < 0.0001$, zero of 10,000 surrogates matching intact performance); Plus Lag-1 (Rank) fell from 0.8591 to 0.7732 ± 0.0416 (range: $[0.6620, 0.9098]$, empirical $p = 0.0178$); and Plus Jump (D_1) degraded from 0.8611 to 0.7783 ± 0.0040 ($p < 0.0001$).

This frozen scaling bounds all 10,000 shuffled linear iterations below 0.8370. Across $M \approx 5{,}133$ adjacent blocks, the shuffled increment D_1 converges to $\mathbb{E}[|X - Y|]$ (variance $\approx 2 \times 10^{-4}$), collapsing to a static marginal quantity that aligns with the canonical baseline (0.8182 ± 0.0049).

E. Comparison with Supervised Machine Learning and Domain Baselines

Evaluating the four-feature suite under identical preprocessing enabled a controlled comparison across decision rules (Table I; Fig. 3b). Equal-weight linear scoring achieved 0.9324 AUROC, compared with 0.8813 for source-tuned ridge, 0.8964 for source-tuned logistic regression, and 0.8769 for shrinkage LDA. Its gain over the strongest matched baseline, logistic regression, was $+0.0360$ (paired 95% CI $[+0.0090, +0.0728]$); gains over ridge and LDA were $+0.0511$ and $+0.0556$, respectively. These comparisons isolate the scoring rule while holding input features and preprocessing constant. Additional domain baselines yielded 0.5136 AUROC for LZC, 0.6953 for spectral entropy, and 0.5702 for the EEGNet-style CNN (Supplementary Table S2).

F. Adaptive Gated Consensus (T-GECF) and Noise Stress Testing

On the complete representation $\mathbf{x}_{\text{combined}}$, continuous Sigmoid volatility gating in T-GECF ($\theta_0 = 1.645, \kappa = 2.0$) achieved an AUROC of **0.9116** (95% CI $[+0.0013, +0.1927]$ vs. canonical baseline; positive gain probability 97.68%) with 0.8399 balanced accuracy, 0.9833 sensitivity, 0.6966 specificity, and PR-AUC = **0.8669** (Table I). On $\mathbf{x}_{+\rho_1}$, T-GECF similarly achieved 0.8828 AUROC ($+0.0237$ over standard rank consensus 0.8591).

To assess resilience against sensor disturbances, impulse spike stress tests introduced outlier perturbations ($\pm\delta\sigma$, $\delta \in \{0, 5, 10, 20\}$) into individual feature coordinates across testing partitions. Under extreme 20σ impulse corruption, unconstrained linear aggregation dropped from 0.8911 to 0.7404 (-0.1507). In contrast, T-GECF shifted weight toward bounded rank consensus, retaining an AUROC of 0.8518 ($+0.1114$ over linear aggregation) while maintaining 0.9116 AUROC under unperturbed conditions.

G. Multiscale Window Sensitivity and Longitudinal Reliability

Evaluating observational windows (Table II-A) confirmed consistent linear discrimination across 2 s (0.8916), 4 s (0.8911), and 8 s (0.8711) partitions, with rank consensus similarly stable (0.8453–0.8591). Truncating recordings showed that a 30-minute record attained 0.8867 AUROC, close to full-record performance. Test-retest reliability across five 1-hour segments ($N = 38$, 190 patient-hours) yielded $\text{ICC}(3, 1) = 0.4857$ for ρ_1, 0.6433 for D_1, and 0.4511 for r_{residual}. Higher-order lags decayed monotonically ($\rho_1 = 0.450 \rightarrow \rho_2 = 0.401 \rightarrow \rho_3 = 0.389$ in MCS; $0.287 \rightarrow 0.252 \rightarrow 0.208$ in VS/UWS), confirming multi-second persistence. Under adjacency masking $\mathbb{I}_{\text{adj}}$, 50-iteration Monte Carlo dropout demonstrated stability ($r = 0.9942 \pm 0.0030$ at 10% dropout; $r = 0.9199 \pm 0.0308$ at 50% loss), while tightening artifact rejection to $150\,\mu\text{V}$ preserved 0.8702 AUROC (+0.1729 over unrejected baseline 0.6973; +0.0502 over unconstrained baseline 0.8200). Contiguous 30-min block removal ($r = 0.9840 \pm 0.0166$) and partition depths (10- to 100-fold CV, 0.8900–0.9002) confirmed stability across analytical scales.

H. External Waveform Validation and Sleep Macro-Dynamics

1) Rapid Bedside Triage (Figshare, $N = 50$): On 120-s records, the jump-rate branch achieved 0.7691 AUROC versus 0.7199 for the static baseline. On identical features, source-tuned ridge, logistic regression, and LDA achieved 0.7759, 0.7674, and 0.7742 (Supplementary Table S2). The equal-weight–ridge difference was -0.0068 (paired 95% CI $[-0.0319, +0.0125]$). At source-derived thresholds, jump-rate scoring attained 1.0000 sensitivity (14/14 MCS detected) with 0.5645 balanced accuracy (specificity 0.1290; Table II-B), prioritizing false-negative exclusion during rapid triage. *2) Cross-Hardware Inpatient Validation (CRRC, $N = 15$)*: Under shared target standardization and direction alignment, the four-feature linear suite achieved 0.7778 AUROC, rising to 0.7963 with fixed $c = 1.5$ truncation; matched ridge, logistic regression, and LDA yielded 0.7407, 0.7593, and 0.7593. In the drift-free subset ($N = 13$), linear scoring, truncated scoring, and ridge each attained 0.7143, while logistic regression and LDA reached 0.6905. Frozen-source transfer yielded 0.5741 on the full cohort, advancing to 0.7778 under target

adaptation (Supplementary Table S2). At source thresholds, full-cohort truncated scoring achieved sensitivity 0.6667 and specificity 0.7778. *3) Circadian Sleep Macro-Dynamics (Wielek, $N = 23$)*: Polysomnography across 24-hour records (11 MCS, 12 VS/UWS) confirmed that temporal continuity extends to macro-scale sleep architecture: VS/UWS patients exhibited state trapping ($P_{\text{stay}} = 0.8361 \pm 0.1001$), whereas MCS retained transition flexibility ($T_{\text{trans}} = 0.261 \pm 0.099$ vs. 0.164 ± 0.100, one-sided $p = 0.0302$, AUROC = 0.7348; nocturnal sleep $p = 0.0392$, AUROC = 0.7576; Table II-B).

I. Hyperparameter Sensitivity and Gating Grid Evaluation

Supplementary Table S3-A reports systematic benchmarks across truncation thresholds c and continuous Sigmoid gating parameters (θ_0, κ). Varying Winsorized soft truncation from $c = 1.0\sigma$ to $c = \infty$ produced stable AUROC within $[0.8958, 0.9149]$ (balanced accuracy 0.8046–0.8399, sensitivity 0.9333–0.9833), peaking at $c = 1.5\sigma$ (0.9149) before approaching unconstrained linear aggregation (0.8996).

Evaluating T-GECF across a 4×4 grid crossing Gaussian quantile thresholds $\theta_0 \in \{1.282, 1.645, 1.960, 2.326\}$ and slopes $\kappa \in \{0.5, 1.0, 2.0, 5.0\}$ demonstrated stability (Supplementary Table S3-B). Across all 16 configurations, AUROC remained bounded within 0.8996–0.9220, exceeding the canonical baseline (0.8200) by $+0.08$ to $+0.10$. At $\theta_0 = 2.326$ ($\kappa = 5.0$), $\beta(\mathbf{x}) \to 0$, smoothly recovering linear aggregation (0.8996).

IV. Discussion

A. Physiological Significance of Autoregressive Memory

Under the mesocircuit hypothesis [19], reciprocal excitatory loops between the central thalamus and frontoparietal cortices sustain contiguous multi-second theta continuity in MCS, reflecting cortical reverberation. Conversely, severe deafferentation in VS/UWS disrupts these pathways, degrading theta fluctuations into decoupled transients that rapidly decay ($\rho_1 \to 0$). Permutation-invariant dispersion ($-\text{CV}_\theta$) computes only marginal variance, collapsing ordered reverberation and noise into identical scores. Enforcing chronological continuity via physical adjacency masking ($\mathbb{I}_{\text{adj}}$) preserves this distinction, yielding an auditable bedside marker of thalamocortical integrity.

These cross-cohort findings link temporal dynamics to distinct recording regimes. Continuous EEG enables asymptotic estimation of ρ_1, whereas the absolute-increment branch (D_1) retains 0.7691 AUROC on short Figshare records. External CRRC evaluation shows that feature alignment and site scaling govern transfer alongside scoring rules. At the circadian scale, elevated sleep-stage transition flexibility in MCS captures transitions between discrete states, supporting scale-specific temporal descriptors paired with explicit calibration.

B. Resolution of Non-Specific Bilateral Confounding

Substituting raw bilateral correlation with residual correlation (r_{residual}, AUROC 0.8433) decouples common-mode activity accounting for $47.27\% \pm 18.43\%$ of raw bilateral variance from volume conduction. Diffuse slow waves project near-synchronously across hemispheres, causing unadjusted correlation to conflate volume conduction with functional connectivity. Decoupling this common mode unmasks genuine differentiation: pathological hyper-synchrony in VS/UWS (0.376 ± 0.193) versus preserved modularity in MCS (0.233 ± 0.080, $p = 0.0211$). In subacute CRRC patients, higher residual correlation in MCS reflects an inverted association direction evaluated under sign alignment ($\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$; Supplementary Note C). In sparse arrays, 4-channel averaging generalizes to subspace projection ($\mathbf{P}^\perp = \mathbf{I} - \mathbf{Z}\mathbf{Z}^+$; 46.52% Mode-1 variance across 62 channels, Table II-B).

C. Operator Dynamics: Rank Consensus vs. Linear Aggregation

Matched internal benchmarks support fixed-weight aggregation as an effective small-cohort scoring rule: the four-feature linear suite retained an AUROC advantage over supervised models trained on identical inputs. External validation revealed comparable performance across classifiers, indicating that feature representation and domain calibration dominate transfer. Soft truncation increased full-cohort CRRC AUROC from 0.7778 to 0.7963, while corresponding discovery AUROC remained 0.9291 (versus 0.9324 unclipped), providing a bounded-influence scoring variant. Adaptive gating via T-GECF complements this by combining linear resolution with rank stability under impulse perturbations.

D. Clinical Scope, Cognitive Motor Dissociation, and Multimodal Triage

A primary clinical application is detecting covert consciousness in cognitive-motor dissociation (CMD), where CRS-R misses volitional responses in up to 25% of unresponsive patients [5]. Continuous lag-1 persistence (ρ_1) provides an auditable marker to prioritize candidates for confirmatory active EEG, fMRI/PET, or TMS-EEG [4], [6]. At discovery prevalence ($12/41$), with 0.9167 sensitivity and 0.7034 specificity at the training-median threshold, the referral model $\Delta_{\text{referral}} = 1 - [P_{\text{MCS}}\,\text{Sens} + (1 - P_{\text{MCS}})(1 - \text{Spec})]$ projects 52.2% fewer transfers than universal escalation before specialized protocols (Supplementary Note C).

E. Comparison with Recent Advanced Paradigms (2024–2026)

Recent dense-array and sensory paradigms report high DoC discrimination: 128-channel microstates reach 0.97 AUC [12], entropy CNNs reach 0.902 AUC [29], multimodal pipelines achieve 92.4% accuracy [30], auditory features differentiate consciousness levels [21], and source graph metrics yield 0.714–0.893 AUC [31]. Translating them to bedside surveillance faces obstacles: 64–128 channel arrays and source imaging are motion-sensitive, acoustic tests require specialized hardware, and deep models risk overfitting ($N < 50$). Combining order-aware persistence (ρ_1, D_1) with common-mode decoupling (r_{residual}) enables 4–6 channel operation with 2.15 ms latency and robust multicenter utility (0.7778–0.7963 on CRRC; 0.7691 on Figshare; $N = 129$).

F. Limitations

Limitations include: (1) discovery in 41 patients (12 MCS), warranting prospective multicenter validation; (2) external cohorts test distinct regimes: subacute telemetry (CRRC) requires unsupervised cross-hardware harmonization, whereas archival triage records (Figshare) assess short-epoch jump dynamics (D_1) rather than continuous memory; (3) test-retest reliability of ρ_1 was moderate (ICC $= 0.4857$), favoring continuous telemetry over brief snapshots; (4) cross-sectional designs necessitate prospective trials to confirm referral reduction; and (5) sample sizes limit statistical resolution for etiology and sedation subgroups.

V. Conclusion

Order-aware temporal descriptors and common-mode-adjusted bilateral dynamics provide a compact EEG representation for bedside consciousness assessment. On identical four-feature inputs, equal-weight consensus achieved 0.9324 discovery AUROC versus 0.8964 for matched supervised logistic regression. External validation yielded 0.7778 AUROC on CRRC (0.7963 with soft truncation) and 0.7691 on short Figshare records. With domain calibration and 2.15 ms latency, these results establish an auditable bedside decision support system for continuous neuromonitoring.

Acknowledgment

OpenAI Codex assisted code implementation, preparation of the data figures and system schematic, and revision of the Methods, Results, and Discussion. The authors are responsible for verification of the code, figures, results, and scientific interpretations.

Declarations

Ethics and Consent: Secondary analyses complied with Declaration of Helsinki, approved by institutional ethics committees: FRCCICMR Moscow [20], Salzburg Ethics Commission (No. 415-E/1517/14-2013) [13], Hangzhou Mingzhou Hospital [12], and China Rehabilitation Research Center (CRRC, Ethics No. 2025-118-01) [21], with written surrogate consent. **Data and Code Availability:** Public data: Mendeley [20] and Figshare [12], [22]. De-identified CRRC tabular feature matrices and replication scripts: `https://github.com/ujujuj3061831150/DoC_Temporal_Dynamics`. Raw clinical telemetry from CRRC is restricted by institutional patient confidentiality regulations. **Conflict of Interest:** The authors declare no competing interests.

References

[1] J. T. Giacino, S. Ashwal, N. Childs, R. Cranford, B. Jennett, D. I. Katz *et al.*, "The minimally conscious state: definition and diagnostic criteria," *Neurology*, vol. 58, no. 3, pp. 349–353, 2002.

[2] J. T. Giacino, D. I. Katz, N. D. Schiff, J. Whyte, E. J. Ashman, S. Ashwal *et al.*, "Practice guideline update recommendations summary: Disorders of consciousness," *Neurology*, vol. 91, no. 10, pp. 450–460, 2018.

[3] C. Schnakers, A. Vanhaudenhuyse, J. Giacino, M. Ventura, M. Boly, S. Majerus, G. Moonen, and S. Laureys, "Diagnostic accuracy of the vegetative and minimally conscious state: clinical consensus versus standardized neurobehavioral assessment," *BMC Neurology*, vol. 9, p. 35, 2009.

[4] D. Cruse, S. Chennu, C. Chatelle, T. A. Bekinschtein, D. Fernández-Espejo, J. D. Pickard, S. Laureys, and A. M. Owen, "Bedside detection of awareness in the vegetative state: a cohort study," *The Lancet*, vol. 378, no. 9809, pp. 2088–2094, 2011.

[5] Y. G. Bodien, J. Allanson, P. Cardone, A. Bonhomme, J. Carmona, C. Chatelle *et al.*, "Cognitive motor dissociation in disorders of consciousness," *New England Journal of Medicine*, vol. 391, no. 7, pp. 598–608, 2024.

[6] A. G. Casali, O. Gosseries, M. Rosanova, M. Boly, S. Sarasso, K. R. Casali, S. Casarotto, M.-A. Bruno, S. Laureys, G. Tononi, and M. Massimini, "A theoretically based index of consciousness independent of sensory processing and behavior," *Science Translational Medicine*, vol. 5, no. 198, p. 198ra105, 2013.

[7] S. Chennu, J. Annen, S. Wannez, A. Thibaut, C. Chatelle, and H. Cassol, "Brain networks predict metabolism, diagnosis and prognosis at the bedside in disorders of consciousness," *Brain*, vol. 140, no. 8, pp. 2120–2132, 2017.

[8] J. D. Sitt, J.-R. King, I. El Karoui, B. Rohaut, F. Faugeras, A. Gramfort, L. Cohen, M. Sigman, S. Dehaene, and L. Naccache, "Large scale screening of neural signatures of consciousness in patients in a vegetative or minimally conscious state," *Brain*, vol. 137, no. 8, pp. 2258–2270, 2014.

[9] D. A. Engemann, F. Raimondo, J.-R. King, B. Rohaut, G. Louppe, F. Faugeras *et al.*, "Robust EEG-based cross-site and cross-protocol classification of states of consciousness," *Brain*, vol. 141, no. 11, pp. 3179–3192, 2018.

[10] C. A. Bareham, N. Roberts, J. Allanson, P. J. A. Hutchinson, J. D. Pickard, D. K. Menon, and S. Chennu, "Bedside EEG predicts longitudinal behavioural changes in disorders of consciousness," *NeuroImage: Clinical*, vol. 28, p. 102372, 2020.

[11] C. Bandt and B. Pompe, "Permutation entropy: a natural complexity measure for time series," *Physical Review Letters*, vol. 88, no. 17, p. 174102, 2002.

[12] Y. Ling, X. Wen, J. Tang, Z. Tao, L. Sun, H. Xin, and B. Luo, "Effect of topographic comparison of electroencephalographic microstates on the diagnosis and prognosis prediction of patients with prolonged disorders of consciousness," *CNS Neuroscience & Therapeutics*, vol. 30, no. 3, p. e14421, 2024.

[13] T. Wielek, J. Lechinger, M. Wislowska, C. Blume, P. Ott, S. Wegenkittl *et al.*, "Sleep in patients with disorders of consciousness characterized by means of machine learning," *PLoS One*, vol. 13, no. 1, p. e0190458, 2018.

[14] J. Xiao, T. Yu, H. Huang, J. Pan, F. Wang, D. Chen *et al.*, "Assessing disorders of consciousness using temporal sleep dynamics extracted from whole-night PSG," *IEEE Transactions on Biomedical Engineering*, pp. 1–14, 2026, early access, doi: 10.1109/TBME.2026.3675367.

[15] T. Thiéry, T. Lajnef, E. Combrisson, A. Dehgan, P. Rainville, G. A. Mashour *et al.*, "Long-range temporal correlations in the brain distinguish conscious wakefulness from induced unconsciousness," *NeuroImage*, vol. 179, pp. 30–39, 2018.

[16] X. Wei, Z. Yan, L. Cai, M. Lu, G. Yi, J. Wang *et al.*, "Aberrant temporal correlations of ongoing oscillations in disorders of consciousness on multiple time scales," *Cognitive Neurodynamics*, vol. 17, no. 3, pp. 633–645, 2023.

[17] Y. Bai, J. He, X. Xia, Y. Wang, Y. Yang, H. Di *et al.*, "Spontaneous transient brain states in EEG source space in disorders of consciousness," *NeuroImage*, vol. 240, p. 118407, 2021.

[18] D. Toker, I. Pappas, J. D. Lendner, J. Frohlich, D. M. Mateos, S. Muthukumaraswamy *et al.*, "Consciousness is supported by near-critical slow cortical electrodynamics," *Proceedings of the National Academy of Sciences*, vol. 119, no. 7, p. e2024455119, 2022.

[19] N. D. Schiff, "Recovery of consciousness after brain injury: a mesocircuit hypothesis," *Trends in Neurosciences*, vol. 33, no. 1, pp. 1–9, 2010.

[20] J. Nekrasova, M. Kanarskii, D. Yankevich, I. Borisov, and P. Pradhan, "Polysomnographic records of patients with disorders of consciousness," Mendeley Data, V1, 2020.

[21] F. Duan, J. Chen, Z. Yan, X. Cao, G. Xiong, and F. Li, "Auditory paradigm-dependent EEG feature screening and representation for distinguishing consciousness states in male patients with disorders of consciousness," *Frontiers in Neurology*, vol. 17, p. 1927313, 2026.

[22] C. Xu, J. Zou, F. He, X. Wen, J. Li, J. Gao *et al.*, "Neural tracking of sound rhythms correlates with diagnosis, severity, and prognosis of disorders of consciousness," *Frontiers in Neuroscience*, vol. 15, p. 646543, 2021.

[23] R. M. Dawes, "The robust beauty of improper linear models in decision making," *American Psychologist*, vol. 34, no. 7, pp. 571–582, 1979.

[24] M. Grabisch, J.-L. Marichal, R. Mesiar, and E. Pap, *Aggregation Functions*. Cambridge University Press, 2009.

[25] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[26] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

[27] D. B. Rubin, "The Bayesian bootstrap," *The Annals of Statistics*, vol. 9, no. 1, pp. 130–134, 1981.

[28] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, and J. D. Farmer, "Testing for nonlinearity in time series: the method of surrogate data," *Physica D: Nonlinear Phenomena*, vol. 58, no. 1-4, pp. 77–94, 1992.

[29] S. Qu, X. Wu, L. Huang, Y. Zhou, Q. Sun, and F. Zeng, "Diagnosis of disorders of consciousness using nonlinear feature derived EEG topographic maps via deep learning," *Scientific Reports*, vol. 16, p. 7417, 2026.

[30] W. Zhang, X. Shi, M. Li, L. Zhang, R. Zhang, X. Wu *et al.*, "Assess the level of consciousness in patients with disorders of consciousness by combining resting-state and auditory-evoked EEG," *Frontiers in Neuroscience*, vol. 19, p. 1613356, 2025.

[31] Z. Hao, X. Xia, Y. Pan, Y. Bai, Y. Wang, B. Peng, and W. Dou, "Uncovering brain network insights for prognosis in disorders of consciousness: EEG source space analysis and brain dynamics," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 144–153, 2024.